1. Hadoop MapReduce

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Ejercicio 1

Código del mapper

```
#!/usr/bin/python

import sys
import re

word = sys.argv[1].lower()

for line in sys.stdin:
        words = re.findall('[a-z]+', line.lower())

        if word in words:
        print(line)
```

Código del reducer

```
#No hay reducer
```

Comando

```
gcloud dataproc jobs submit hadoop --cluster=micluster \
--jar=file:///usr/lib/hadoop/hadoop-streaming.jar -- -files \
gs://$GOOGLE_CLOUD_PROJECT/t1/mapper1.py \
-mapper "mapper1.py zoroaster" -reducer NONE \
-input gs://cloudandbigdata/Gutenberg -output gs://$GOOGLE_CLOUD_PROJECT/t1/ej1
```

Salida de `gcloud storage cat gs://$GOOGLE_CLOUD_PROJECT/t1/ej1/* | head`

```
The Magus Zoroaster, my dead child,

To-night the triple Zoroaster

always will be. The Indian Vedas, the Nackas of Zoroaster, the Tal

[664] ["For the creed of Zoroaster," see Sir Walter Scott, _Letters on

bystanders when Zoroaster died.  For whatever is truly wondrous
```

Ejercicio 2

Primer trabajo

Código del mapper

```
#!/usr/bin/python

import sys

for line in sys.stdin:
   tokens = line.split()

   for i in range(len(tokens)):
      if tokens[i] == "\"GET":
         print(tokens[i+1].split('?')[0] + "\t1")
```

Código del reducer

```
#!/usr/bin/python

import sys
prev = ""
count = 0

for line in sys.stdin:
   key, value = line.split('\t')

   if key == prev:
      count += int(value)
   else:
      if prev != "":
         print(prev + '\t' + str(count))
      prev = key
      count = int(value)

if prev != "":
   print(prev + '\t' + str(count))
```

Comando

```
gcloud dataproc jobs submit hadoop --cluster=micluster \
--jar=file:///usr/lib/hadoop/hadoop-streaming.jar -- -files \
gs://GOOGLE_CLOUD_PROJECT/t1/mapper2a.py, gs: //GOOGLE_CLOUD_PROJECT/t1/reducer2a.py
\
-mapper mapper2a.py -reducer reducer2a.py \
-input gs://cloudandbigdata/access_log -output gs://$GOOGLE_CLOUD_PROJECT/t1/ej2tmp
```

Salida de `gcloud storage cat gs://$GOOGLE_CLOUD_PROJECT/t1/ej2tmp/* | head`

```
/#tabs-2          7
/%22.php          1
/*.detail.php   3
/////displaytitle.php   1
/////email.php  1
/////productionnotes.php          1
////displaytitle.php        1
///displaytitle.php         1
///index.php      9
///productionnotes.php  1
```

Segundo trabajo

Código del mapper

```
#!/usr/bin/python

import sys

list = []

for line in sys.stdin:
   info = line.split('\t')
   list.append((info[0], int(info[1])))
   list.sort(key= lambda x : x[1], reverse= True)
   list = list[:100]

for element in list:
   print(f"{element[0]}\t{element[1]}")
```

Código del reducer

```
#!/usr/bin/python

import sys

list = []

for line in sys.stdin:
   info = line.split('\t')
   list.append((info[0], int(info[1])))
   list.sort(key= lambda x : x[1], reverse= True)
   list = list[:100]

for element in list:
   print(f"{element[0]}\t{element[1]}")
```

Comando

```
gcloud dataproc jobs submit hadoop --cluster=micluster \
--jar=file:///usr/lib/hadoop/hadoop-streaming.jar -- \
-D mapreduce.job.reduces=1 \
-files gs://$GOOGLE_CLOUD_PROJECT/t1/mapper2b.py \
-mapper mapper2b.py -reducer mapper2b.py \
-input gs://$GOOGLE_CLOUD_PROJECT/t1/ej2tmp -output gs://$GOOGLE_CLOUD_PROJECT/t1/ej2
```

Salida de `gcloud storage cat gs://$GOOGLE_CLOUD_PROJECT/t1/ej2/* | head`

```
/displaytitle.php           261841
/downloadSingle.php    138919
/assets/css/combined.css       117326
/assets/js/javascript_combined.js      106974
/           99795
/assets/img/home-logo.png      98746
/assets/css/printstyles.css    93151
/images/filmpics/0000/3695/Pelican_Blood_2D_Pack.jpg           91884
/favicon.ico       66829
/release-schedule/         62479
```

Ejercicio 3

Código del mapper

```
#!/usr/bin/python

import sys

is_header = True

for line in sys.stdin:
   if is_header:
      is_header = False
      continue
   words = line.strip().split(',')
   print(f"{words[1]}\t({words[3]}, {words[3]}, {1})")
```

Código del reducer

```
#!/usr/bin/python

import sys

prev = ""
min, max, total = float('inf'), 0, 0

for line in sys.stdin:
   id, value = line.strip().split('\t')
   value = value.strip().strip('()')
   aux1, aux2, aux3 = value.strip().split(',')
   minDate, maxDate, count = int(aux1), int(aux2), int(aux3)

   if id == prev:
      if minDate < min: min = minDate
      if maxDate > max: max = maxDate
      total += count
   else:
      if prev != "":
         print(f"{prev}\t({min}, {max}, {total})")
      prev = id
      min, max, total = minDate, maxDate, count

if prev != "":
         print(f"{prev}\t({min}, {max}, {total})")
```

Comando

```
gcloud dataproc jobs submit hadoop --cluster=micluster \
--jar=file:///usr/lib/hadoop/hadoop-streaming.jar -- -files \
gs://GOOGLE_CLOUD_PROJECT/t1/mapper3.py, gs://GOOGLE_CLOUD_PROJECT/t1/reducer3.py \
-mapper mapper3.py -reducer reducer3.py \
-input gs://cloudandbigdata/Kindle_Store.csv -output gs://$GOOGLE_CLOUD_PROJECT/t1/ej3
```

Salida de `gcloud storage cat gs://$GOOGLE_CLOUD_PROJECT/t1/ej3/* | head`

```
B000FA5PV4    (1264983448000, 1535426541437, 22)
B000FA5QH2    (1198770082000, 1680742988778, 187)
B000FA5QMM    (1301013737000, 1680508606666, 621)
B000FA5SCK    (1308952315000, 1627525747048, 63)
B000FA5SES    (1345521138000, 1637878810980, 9)
B000FA5SUM    (1270996202000, 1682084828562, 315)
B000FA6680    (1286647725000, 1518868001513, 13)
B000FA66SA    (1391920468000, 1622376458543, 8)
B000FA66TE    (1351646816000, 1594867141039, 24)
B000FA66UI    (1285792036000, 1673085654636, 7)
```

Ejercicio 4

Primer trabajo

Código del mapper

```
#!/usr/bin/python

import sys

is_header = True

for line in sys.stdin: #acumulado, num, id
   if is_header:
      is_header = False
      continue
   words = line.strip().split(',')
   print(f"{words[2]}\t{1}\t{words[1]}")
```

Código del reducer

```
#!/usr/bin/python

import sys

prev = ""
sum, count = 0, 0

for line in sys.stdin:
   aux1, aux2, id = line.strip().split('\t')
   rating, number = float(aux1), float(aux2)

   if id == prev:
      sum += rating * number
      count += number
   else:
      if prev != "":
         print(f"{sum/count:.1f}\t{count}\t{prev}")
      prev = id
      sum, count = rating * number, number

if prev != "":
   print(f"{sum/count:.1f}\t{count}\t{prev}")
```

Comando

gcloud dataproc jobs submit hadoop --cluster=micluster \
--jar=file:///usr/lib/hadoop/hadoop-streaming.jar -- -files \
gs://$GOOGLE_CLOUD_PROJECT/t1/mapper4a.py, gs://GOOGLE_CLOUD_PROJECT/t1/reducer4a.py \
\
-mapper mapper4a.py -reducer reducer4a.py \
-input gs://cloudandbigdata/ml-latest/ratings.csv -output
gs://$GOOGLE_CLOUD_PROJECT/t1/ej4tmp

Salida de `gcloud storage cat gs://$GOOGLE_CLOUD_PROJECT/t1/ej4tmp/* | head`

```
3.0     1.0     159415
3.0     1.0     3359
3.0     1.0     81847
3.0     1.0     6294
3.0     1.0     6320
3.0     1.0     71282
3.0     1.0     1198
3.0     1.0     1127
3.0     1.0     3358
3.0     1.0   166621
```

Segundo trabajo

Código del mapper

```
#Usamos la funcion cat
```

Código del reducer

```
#!/usr/bin/python

import sys

prev = ""
ratingList = []

for line in sys.stdin:
   mean, count, id = line.strip().split('\t')

   if mean == prev:
      ratingList.append(id)
   else:
      if prev != "":
         print(f"{prev}\t{ratingList}")
      prev = mean
      ratingList = []

if prev != "":
         print(f"{prev}\t{ratingList}")
```

Comando

```
gcloud dataproc jobs submit hadoop --cluster=micluster \
--jar=file:///usr/lib/hadoop/hadoop-streaming.jar -- -files \
gs://$GOOGLE_CLOUD_PROJECT/t1/reducer4b.py \
-mapper cat -reducer reducer4b.py \
-input gs://GOOGLE_CLOUD_PROJECT/t1/ej4tmp − outputgs: //GOOGLE_CLOUD_PROJECT/t1/ej4
```

Salida de `gcloud storage cat gs://$GOOGLE_CLOUD_PROJECT/t1/ej4/* | head | cut -c -80`

```
3.0     ['92058', '434', '339', '316', '296', '33493', '293', '292', '91653', '34048
4.5     ['109374', '2571', '551', '1270', '6333', '58559', '54286', '471', '208', '6
1.5     ['113345', '6380', '7373', '74946', '81834', '69526', '114935', '8917', '531
4.0     ['2571', '2115', '1097', '1101', '1210', '1206', '1196', '858', '1204', '778
1.0     ['4142', '87', '1917', '714', '5009', '231', '586', '5378', '5', '4995', '17
2.5     ['1975', '68954', '4207', '8376', '5900', '41569', '119966', '152591', '8866
5.0     ['1198', '1214', '2502', '2959', '4011', '4027', '4033', '5618', '8874', '48
2.0     ['2808', '87520', '627', '1918', '45517', '1235', '2806', '158398', '4370',
3.5     ['6952', '6950', '104913', '77837', '3034', '79132', '3019', '736', '81229',
0.5     ['6548', '48385', '5737', '3821', '81782', '1777', '6711', '5736', '80549',
```